1. A coin is to be tossed four times. Define events X and Y such that

   - X: first and last coin have opposite faces
   - Y: exactly two heads appear.

   Assume that each of the sixteen head/tail sequences has the same probability. Compute $P(X^C \cap Y)$. Write a decimal number with three decimal places.

   Solution:

   $$X^C \cap Y = \{(H,T,T,H),(T,H,H,T)\}$$

   then

   $$P(X^C \cap Y) = \frac{2}{16} = 0.125.$$

2. One hundred voters were asked their opinions on two candidates, A and B, running for political seat for the federal parliament. Their responses to three questions are summarised as following: 65 respondents said they like candidate A, 55 said they like candidate B, and 25 said they like both candidates. Of the voters who like at least one candidate, what proportion likes both? Write a decimal number with three decimal places.

   Solution:

   $$P(A \cap B | A \cup B) = \frac{P(A \cap B)}{P(A \cup B)} = \frac{\frac{25}{100}}{\frac{55+65-25}{100}} = \frac{25}{95} = 0.263$$

   because $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

3. How many probability equations need to be verified to establish the mutual independence of four events?

   Solution: We have 6 verifications of form $P(A_i \cap A_j) = P(A_i)P(A_j)$, then 4 verifications of form $P(A_i \cap A_j \cap A_k) = P(A_i)P(A_j)P(A_k)$ and 1 verification of form $P(A_i \cap A_j \cap A_k \cap A_\ell) = P(A_i)P(A_j)P(A_k)P(A_\ell)$. So we need 11 verifications in total.

4. Nine students, five men and four women, interview for four summer internships sponsored by a city newspaper. In how many ways can the newspaper choose a set of four interns if it must include two men and two women in each set? Write an integer.

   Solution:

   $$\binom{5}{2} \times \binom{4}{2} = 60.$$

5. An urn contains six chips, numbered 1 through 6. Two are chosen at random and their numbers are added together. What is the probability that the resulting sum is equal to 5? (Consider that sampling is performed without replacement and initially all chips have the same probability to be sampled). Write a decimal number with three decimal places.

   Solution: There are two cases in which the sum can be equal to 5: $\{(4,1),(2,3)\}$. Then

$$\Pr(\text{sum} = 5) = \frac{\text{number of pairs summing to 5}}{\text{number of pairs}} = \frac{2}{\binom{6}{2}} = \frac{2}{15} = 0.1333333.$$

6. If a family has four children, it is more likely they will have

   - two boys and two girls (OPTION A) or
   - three of one sex and one of another (OPTION B)?

   Assume that the probability of a child being a boy is 0.5 and that the births are independent events.

   Solution:

   Define X as the number of girls. Then $X \sim Bin(4, 0.5)$. Option A has probability

$$P(X = 2) = \binom{4}{2} 0.5^2 0.5^2 = 0.375,$$

   while Option B has probability

$$\begin{aligned} P(X = 1) &= \binom{4}{1} 0.5^1 0.5^3 = 0.25 \\ P(X = 3) &= \binom{4}{3} 0.5^3 0.5^3 = 0.25 \\ P(X = 1) + P(X = 3) &= 0.25 + 0.25 = 0.5. \end{aligned}$$

   Then Option B is more likely.

7. Brett and Margo have each thought about murdering their rich uncle Basil in hopes of claiming their inheritance. Hoping to take advantage of Basil's predilection for dessert, Brett has put rat poison into the cherries flambé; Margo, unaware of Brett's activities, has laced the chocolate mousse with cyanide. Given the amount likely to be eaten, the probability of the rat poison being fatal is 0.60 and is 0.90 for the cyanide. Based on past dinners, we can assume that Basil has 50% chance of asking for the cherries flambé, 40% chance of asking for the chocolate mousse, and 10% chance of skipping dessert. As soon as dishes

are cleared, Basil drops dead. In the absence of any other evidence, who should be considered the prime suspect?

Solution: Define the events

- B: Basil dies
- A_1: Brett is the murderer, i.e. Basil ate the cherry flambé
- A_2: Margo is the murderer, i.e. Basil ate the chocolate mousse
- A_3: there is no murdered, i.e. Basil did not eat dessert.

Then

$$P(A_1|B) = \frac{P(B|A_1)P(A_1)}{P(B|A_1)P(A_1) + P(B|A_2)P(A_2) + P(B|A_3)P(A_3)} = \frac{0.6 \cdot 0.5}{0.6 \cdot 0.5 + 0.9 \cdot 0.4 + 0.0 \cdot 0.1} = 0.45$$
$$P(A_2|B) = \frac{P(B|A_2)P(A_2)}{P(B|A_1)P(A_1) + P(B|A_2)P(A_2) + P(B|A_3)P(A_3)} = \frac{0.9 \cdot 0.4}{0.6 \cdot 0.5 + 0.9 \cdot 0.4 + 0.0 \cdot 0.1} = 0.55$$
$$P(A_3|B) = \frac{P(B|A_3)P(A_3)}{P(B|A_1)P(A_1) + P(B|A_2)P(A_2) + P(B|A_3)P(A_3)} = \frac{0.0 \cdot 0.1}{0.6 \cdot 0.5 + 0.9 \cdot 0.4 + 0.0 \cdot 0.1} = 0.00$$

Then A_2 is the most likely event, given that B has happened, therefore Margo is the prime suspect.

8. Let E and F be any two independent events with $0 < P(E) < 1$ and $0 < P(F) < 1$. Which of the following statements is NOT true.

- P(exactly one of E and F occurs) = P(E) + P(F) - P(E)P(F)
- P(neither E nor F occurs) = (P(E)-1)(P(F)-1)
- P(E occurs but F does not occur) = P(E) - P(E)P(F)
- P(E occurs given that F does not occur) = P(E)

Solutions:

- P(exactly one of E and F occurs) is not P(E) + P(F) - P(E)P(F) which represents that both events occur
- P(neither E nor F occurs) is $P(E^C \cap F^C)$ and

  $$P(E^C)P(F^C) = [1-P(E)][1-P(F)] = 1-P(F)-P(E)+P(E)P(F) = [P(E)-1][P(F)-1]$$

  so the statement is true
- P(E occurs but F does not occur) is $P(E \cap F^C)$

  $$P(E)P(F^C) = P(E)[1 - P(F)] = P(E) - P(E)P(F) = P(E) - P(E \cap F)$$

  so the statement is true

- P(E occurs given that F does not occur) is $P(E|F^C)$ and

$$P(E|F^C) = P(E)$$

so the statement is true

9. An integer N is to be selected at random from $\{1, 2, \ldots, (10)^3\}$, in the sense that each integer has the same probability of being selected. What is the probability that N will be divisible by 3? Write a number with three decimal places.

Solutions:

$$P(\text{divisible by } 3) = \frac{333}{1000} = 0.333.$$

10. A jeweller has scheduled two appointments to sell engagement rings. Their first appointment will lead to a sale with probability 0.3, and their second appointment will lead independently to a sale with probability 0.6. Any sale made is equally likely to be either for a fancy ring, which cost \$5,000, or a standard ring, which costs \$2,000. Define X as the total dollar value of all sales. What is $p_X(2,000)$, the probability that the jeweller obtain earns \$2,000? Write a number with three decimal places.

Solutions: There are two possible situations that lead to the jeweller to earn \$2,000: either they make a sale for a standard ring in the first appointment or in the second appointment.

Then

$$[P(\text{sale, no sale}) + P(\text{no sale, sale})] \times \frac{1}{2} = [0.3 \cdot 0.4 + 0.7 \cdot 0.6] \times 0.5 = 0.27.$$